International Islamic University Chittagong

Morality Development Program

Semester End Examination, Spring-2025

3rd Semester ***(Other than Shari'ah faculty)***

Course Code: MDP-2302 **Course Title:** ***Tajweedul Qur'an*-II**

Full Marks: 50 Time: 2 Hours &30 Minutes

Answer the following questions
(All questions are of equal value):

1.	Write the meaning of the following Surahs: **(Any two)** a) *Surah Al-Qari'ah (سورة القارعة);* b) *Surah Az-Ziljal (سورة الزلزال);* c) *Surah At-Teen (سورة التين).*	5+5
2.	What is *Lahn*? How many types of *Lahn* are there? Explain each of them briefly with examples.	10
3.	Define *Sifaatul Huruf* (Characteristics of letters). How many types of *Siffaatul Huruf* are there in *Tajweed* primarily and totally? Explain.	10
4.	a. How many types of *Sifatul Huruf Al-Lajimah* are there with opposite? Explain them mentioning their letters. **Or,** b. How many types of *Sifatul Huruf Al-Lajimah* are there without opposite? Explain them mentioning their letters.	10
5.	a. Define *Tafkheem* (Velarization) and *Tarqeek* (Attenuation). How many letters of *Tafkheem* and *Tarqeek* are there in *Tajweed*? Explain. **Or,** b. Explain the rules of Velarization and Attenuation in the letter of *Laam* (ل) of Allah (الله) with examples.	10

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International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination, Spring-2025

Course Code: URED-2302 (URED-2101 for LLB)
Course Title: Sciences of Qur'an and Hadith

Full Marks: 50 **Time: 2 Hours 30 Minutes**

(Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question):

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	Define *Asbab An-Nuzul*. How many types of *Asbab An-Nuzul* are there? Explain some types with examples and benefits.	10	3	Remember & Create
2	Explain how the holy Qur'an was preserved during the lifetime of the Prophet Muhammad (SAAS). Summarize the main reasons for not compiling the holy Qur'an during his lifetime.	10	3	Create
3	"Some inventions of modern science prove the *Ijaz al-Qur'an* (miraculous nature of the holy Qur'an)"- explain elaborately.	10	3	Create
4	Explain the importance and position of Hadith in Islamic Shari`ah.	10	3	Create
5	a. Define Hadith literally and terminologically. Explain some types of Hadith according to the memory and reliability of reporters with examples. **Or,** b. Explain with examples: • *Sanad & Matan;* • *Al-Hadith Al-Qudsi;* • *Al-Hadith Al-Mutawatir;* • *Al-Hadith Al-Garib;* • *Al-Kutub As-Sittah* (Six books of Hadith).	10	3	Understand & Create

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International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE, Semester Final Examination, Spring-2025

Course Code: CHEM- 2301, Course Title: Chemistry

Total marks: *50* Time: *2* hours *30* minutes

[Answer ***all*** the questions. Figures in the right hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Group-A

			Marks
1.	a)	Distinguish between ionization and dissociation. Mention some roles of electrolytes in the body.	2+ 2=4
1.	b)	State Faraday's first law of electrolysis and prove that $m = ZIt$.	3
1.	c)	In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 g and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.	3
2.	a)	State Raoult's law & discuss application of emulsion.	1+2=3
2.	b)	Show that lowering of vapor pressure is a colligative property.	2
2.	c)	State and explain Nernst's Distribution Law with its limitations & application.	1+4= 5

OR

2.	a)	Define specific conductance and molar conductance.	2
2.	b)	Classify the following electrolytes as strong or weak electrolytes: HClO4, H2SO4, NaOH, CH3COOH.	2
2.	c)	What is transport number? Prove that, $t-= 1/1+r$, where t is the transport number of the anion and r is the speed ratio of the ions.	1+2=3
2.	d)	The resistance of decinormal solution of a salt occupying a volume between two platinum electrodes 1.80 cm apart and 5.4 cm^2 in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.	3

Group-B

3.	a)	State rate law and define order of reaction.	2
3.	b)	Define Zero order reaction and Pseudo-unimolecular reaction with appropriate examples.	2+2=4
3.	c)	Define second order reaction. Briefly explain integrated equation for second order reaction and prove that, $k = \frac{1}{t}\frac{x}{a(a-x)}$, where symbols have their usual meanings.	1+3=4
4.	a)	Define homogenous and heterogeneous equilibrium with examples.	2

4. b) State the law of mass action. Derive equilibrium constant (KC & KP) for the following reaction: 1+2=3

$$A + B \rightleftharpoons C + D$$

4. c) Prove that $Kp = Kc \times (RT)^{\Delta n}$ for the following chemical reaction when it is in equilibrium State: 5

$$aA + bB = cC + dD$$

5. a) Define colloids. Mention four colloids based on dispersed phase and dispersion medium. 1+2=3
5. b) What are the differences between lyophilic sol and lyophobic sol? 2
5. c) Define adsorption & write the important assumptions in Langmuir adsorption isotherm. 1+1.5=2.5
5. d) State the Feundlich Adsorption isotherm and show graphical expression for $log\frac{w}{m}$ against $log\ P$. 2.5

OR

5. a) Differentiate between reversible and irreversible reaction. 2
5. b) Discuss briefly characteristics of chemical equilibrium. 3
5. c) Some nitrogen and hydrogen gases are pumped into an empty five-litre glass bulb at 500°C. When equilibrium is established, 3.00 moles of N_2, 2.10 moles of $H2$, and 0.298 mole of $NH3$, are found to be present. Find the value of Kc, for the reaction $N2(g) + 3H2(g) \rightleftharpoons 2NH3(g)$ at 500°C. 3
5. d) Discuss maximum sulphuric acid production by contact process in industry. 2

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE
Semester Final Exam, Spring 2025

Course Code: STAT 2311 | Course Title: Probability and Statistics
Time: 2 hours 30 minutes | Full Marks: 50

(i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Part A

[Answer the questions from the followings]

1. a) Explain the concept of correlation with a suitable example. What is the role of correlation in data analysis? CO2 U 4

Or,

a) Explain the terms regression and regression coefficients. Write dow the properties of regression coefficients. CO2 U 4

1 b) A mobile app development team tracks weekly app crash reports and the number of bug fixes deployed: CO2 E 6

Week	1	2	3	4	5	6	7
Bug Fixes	4	6	3	5	7	6	2
Crash Reports	10	15	8	12	16	13	5

(i) Calculate the correlation coefficient between bug fixes and crash reports. Interpret the result.
(ii) Draw a scatter diagram to represent the above information.

Or,

b) You are testing the performance of a data compression algorithm. The input file size (in MB) is denoted by **x**, and the corresponding compression time (in seconds) is recorded as y. The following data has been collected: CO2 E 6

x	50	60	70	80	90
y	5	6	8	9	11

(i) Determine the linear regression equation that best fits the data.
(ii) Predict the compression time required for a file size of 100 MB.

2 a) Explain the concept of classical probability and axiomatic probability. CO3 R 4

b) A company has three servers- Server A, Server B, and Server C handling 35%, 42%, and 23% of the total requests respectively. Server A fails 2% of the time, Server B fails 1%, and Server C fails 3%. A request is found to have failed. What is the probability that it was handled by: (i) Server A? (ii) Server C? CO3 AP 6

Part B

[Answer the questions from the followings]

3. a) Define random variable with an example. Distinguish between a probability mass function and a probability density function. CO3 U 4

Or

a)	What is mathematical expectation? State two properties of expectation. Give two examples each of discrete and continuous random variables in computing.	CO3	U	4
3. b)	Let X be a continuous random variable representing the time (in seconds) a server takes to respond to a request, modeled by the probability density function: $$f(x) = C(3-x);\ 0 < x < 3$$ (i) Determine the value of the constant C (ii) Calculate the expected response time $E(X)$ (iii) Find the probability that the server takes more than 2 seconds to respond.	CO3	E	6

Or,

b)	In a particular area, the amount of daily rainfall (measured in inches) is modeled by a continuous random variable X with a given probability density function f(x). $$f(x) = \frac{3}{12}\left(6x - 3x^2\right) \quad ,0 < x < 2$$ (i) What is the probability that the rainfall on a randomly chosen day does not exceed 1.6 inches? (ii) What is the probability that the rainfall lies between 0.4 and 1.7 inches on a given day? (iii) Also, determine the expected daily rainfall (mean value in inches) for this region.	CO3	E	6
4. a)	Define the Binomial distribution. What are its key assumptions? Give three practical examples in computing or networking where the Poisson distribution is suitable.	CO3	U	4
b)	A server experiences random requests every second. Suppose the number of requests per second follows a Poisson distribution with λ/m = 3. (i) What is the probability of receiving exactly 5 requests in a second? (ii) What is the probability of receiving at most 2 requests?	CO3	E	6
5. a)	Define p-value and explain how it is used to draw conclusions. How is the t-test applied in hypothesis testing?	CO4	U	4
b)	A CSE lab is testing two compiler versions, Compiler A and Compiler B, to check whether the success of code execution is dependent on the compiler used. The following data was observed from 300 random code submissions:	CO4	C	6

Execution Outcome	Compiler A	Compiler B
Compiled Successfully	90	60
Compilation Error	80	70

At a significance level of $\alpha = 0.05$, test whether the compiler version and compilation success are independent.

Note: Tabulated value of Chi-square value at 1 degree of freedom is 3.84

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE Final Examination, Spring- 2025

Course Title: Mathematics-III Course Code: MATH-2307 (New)

Course Title: Mathematics-IV Course Code: MATH-2401 (Old)

Time: 2 Hours 30 Minutes **Full Marks: 50**

(i) The figures in the right-hand margin indicate full marks

(ii) Course Learning Outcomes and Bloom's Levels are mentioned in additional Columns

Course Learning Outcomes (COs) of the Questions	
CLO1	**Understand** the fundamentals of Matrix , Linear system of equations & Vector analysis
CLO2	**Implement** the fundamental knowledge of Matrix, linear system of equations, vector functions, vector field, scalar field, gradient, divergence, curl, differentiation and integration of vector valued functions, partial derivatives in different problems
CLO3	**Solve** line integrals, surface area, surface integrals, volume integrals, and the work done in different problems
CLO4	**Apply** Green's theorem, Stoke's theorem, Gauss' theorem in solving mathematical problems

Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**App**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

Answer the following questions

1.		Solve the following linear system of equations using LU decomposition method: $x + 2y + 3z = 5, 2x - 4y + 6z = 18$, $3x - 9y - 3z = 6$	**CLO2**	**An**	**10**
2.	**a)**	Suppose that over a certain region of space the electrical potential V is given by $\phi(x,y,z) = 5x^2 - 3xy + xyz$ Find the rate of change (derivative) of the potential at **P(3,4,3)** in the direction of the vector $\vec{v} = \hat{i} + \hat{j} - \hat{k}$	**CLO2**	**U**	**5**
		Or			
		Find all vectors $\vec{v}$ such that $(\hat{i} + 2\hat{j} + \hat{k}) \times \vec{v} = 3\hat{i} + \hat{j} - 5\hat{k}$	**CLO2**	**U**	**5**
2.	**b)**	i) On which condition of λ the vectors $\hat{i} + 2\hat{j} + 3\hat{k}$, $\lambda\hat{i} + 4\hat{j} + 7\hat{k}$, $-3\hat{i} - 2\hat{j} - 5\hat{k}$ are collinear	**CLO1**	U	**2**
		ii) Show that $\vec{A} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{B} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{C} = 3\hat{i} + \hat{j} - \hat{k}$ are coplanar	**CLO1**	**U**	**3**
		Or			
2.	**(b)**	If $\vec{A} = 2\hat{i} - \hat{j} + \hat{k}, \vec{B} = \hat{i} + 3\hat{j} - 2\hat{k}, \vec{C} = -2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{D} = 3\hat{i} + 2\hat{j} + 5\hat{k}$, then find the scalars a, b, c such that $\vec{D} = a\vec{A} + b\vec{B} + c\vec{C}$	**CLO1**	**U**	**5**

Part B

Answer the following questions

3. a) What do you mean by Gradient, Divergence and Curl. Show that $\nabla^2\left(\frac{1}{r}\right) = 0$ **CLO2 R&U 5**

Or

Find the values of constants a, b and c so that, $\vec{F} = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational **CLO2 R&U 5**

b) If $\vec{F} = xy\hat{i} - z\hat{j} + x^2\hat{k}$ and C is the curve curve, $x = t^2$, $y = 2t$, $z = t^3$ from $t = 0$ to $t = 1$, then evaluate the line integral, $\int_c \vec{F} \times d\vec{r}$ **CLO3 App 5**

Or

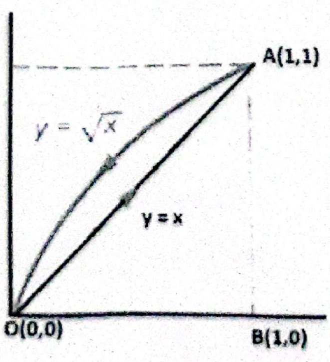
Find the work done in moving a particle once around a circle C in the xy plane, If the circle has center at the origin and radius 3 and the force field is given by $\vec{F} = (2x - y + z)\hat{i} + (x + y - z^2)\hat{j} + (3x - 2y + 4z)\hat{k}$ **CLO3 App 5**

4. **If $\vec{F} = 2y\hat{i} - z\hat{j} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$. Evaluate $\iint_s \vec{F}.\hat{\eta}\, ds$** **CLO3 App 10**

5. Verify the divergence theorem for the vector field $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the region bounded by the planes $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ **CLO4 App 10**

Or

Verify Green's theorem for the integral $\int_C (2xy - x^2)dx + (x + y^2)dy$ taken round the boundary curve c defined by the following graph of the functions $y = x, y^2 = x$ **CLO4 App 10**



Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Final Examination, Spring 2025

Course Code: CSE 2321 **Course Title: Data Structures**

Total marks: *50* *Time: 2 hours 30 minutes*

[Answer *all* the following questions. Figures in the right margin indicate full marks (M). Use a separate answer script for Group-A and Group-B.]

Group A

		M	CO	DL
1. a)	Suppose the following stack of integers is in memory where STACK is allocated N=6 memory cells: TOP=4 STACK: 227,123, N,583,__, __ Find the output of the following program segment: 1. Call POP(STACK, ITEMA). Call POP(STACK, ITEMB). Call POP(STACK, ITEMC). Call PUSH(STACK, ITEMA+140). Call PUSH(STACK, 190). Call PUSH(STACK, ITEMB+ ITEMC). Call PUSH(STACK, ITEMA+ITEMB) Call PUSH(STACK, ITEMA+N) Call PUSH(STACK, ITEMC+N) 2. Repeat while TOP != 0 Call POP(STACK, ITEM). Write: ITEM. 3. Exit. *(N is the last three digits of our ID, i.e. if your ID is C151216, then N=216).*	3	CO1	C2
OR	What is *stack*? Given the following sequence of stack operations on an initially empty stack of size 5, show the content of the stack and value of `TOP` after each operation: `PUSH(11), PUSH(7), POP(), PUSH(9), PUSH(4), PUSH(2)` Then perform complete `POP` operations until the stack is empty and list the popped values.			
b)	Evaluate the following postfix expression using the algorithm that you have studied. P_1: 12, 7, 3 , - , / , 2 , N , 5 , + , * , + Note: Here, N is the last digit of your ID	3	CO2	C3
OR	Write an algorithm to verify whether a given expression has *balanced brackets*. Show the working for: `([{}])`, `{()}`, and `(([])}`.			
c)	Translate the following infix expressions into its equivalent postfix expression using the algorithm that you have studied. **K + L - M*N + (O^P) * W/U/V * T + Q**	4	CO2	C2
2. a)	What is a *circular queue*? Write a procedure to *insert* an element into a circular queue and *delete* an element from a circular queue. What is the time complexity of these operations?	4	CO1	C2
OR	What is *priority queue*? Suppose a *priority queue* is maintained by a *one way list* or *two-dimensional array*. i) Write a procedure which adds an **ITEM** with priority number **M** to the queue. ii) Write a procedure which *removes* an element from the queue and assigns the element to the variable **ITEM**. What is the time complexity of these operations?			
b)	Compare the *one-way list* and *array* representations of priority queues.	1	CO1	C2

c) Consider a recursive function: **3 CO2 C2**

```
int F(int n) {
    if (n <= 3) return 2 * n;
    return F(n - 1) + F(n - 2);
}
```

i) What is the base case?
ii) Calculate `F(2)` and `F(5)`
iii) What type of problem does this recursion model?

d) Define the *Ackermann function* and compute `A(2, 2)`. **2 CO2 C2**

Group B

3. a) Apply selection sort on the array: `{24, 19, 5, 42, 31, 17, 33}`. Show all passes. **3 CO2 C2**

OR Suppose the following numbers are stored in an array A:

`32, XX, 49, 68, 92, 45, 54, 105, 12, 88, 64, 39`

Apply ***merge sort*** algorithm to sort the array A and show each pass separately.
[Here XX is the last two digits of your ID]

b) Which sorting algorithm would you recommend when *swap* operations are highly expensive? Justify your answer based on the following options: **1 CO3 C4**
Heap Sort and Merge Sort
OR Selection Sort and Quick Sort

c) Analysis the complexity of binary search algorithm. **3 CO2 C2**

d) Insert values `{28, 53, 78, 91, 122, 133, 175, 211}` into a hash table of size `11` using the division *method* and *linear probing*. Show the resulting table. **3 CO1 C2**

4. a) Construct an *expression tree* for: **3 CO1 C2**

`((P + Q) * (R - S)) - (T / U + V)`

Then write its *inorder*, *preorder*, and *postorder* traversals.

b) What is a *binary search tree (BST)*? The following preorder traversal corresponds to a BST: **3 CO2 C2**
`40, 20, 10, 30, 70, 60, 90`
Construct the BST and show steps.

OR Explain the three case of deletion operation in binary search tree with necessary figures.

c) What do you mean by *max heap*? Build a max heap from the array: **3 CO1 C2**
[18, 12, 25, 7, 30, 42, 11, 9]. Show the heap at each step.

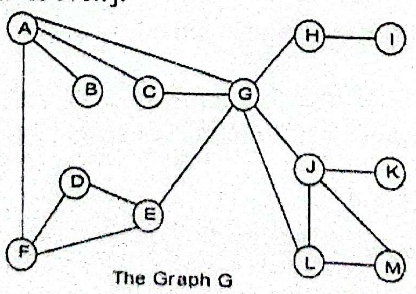
d) To find the top **5** largest salaries from a dataset of **500K** employees, which data structure is most efficient and why? **1 CO3 C4**
i) Max heap ii) BST iii) Min heap iv) Sorted list

5. a) Draw the graph **G** with: **3 CO1 C2**
G = (V, E)
V(G) = {1, 2, 3, 4, 5, 6}
E(G) = {(1,2), (2, 3), (3, 4), (5,1), (5, 6), (2, 6), (1, 6), (4, 6), (2, 4)}
Find the *adjacency matrix* (A) and the *adjacency list.*

b) Use *Warshall's algorithm* to find the *path matrix* for the graph **G** in **Q 5a**. Show each step. **3 CO2 C3**

OR Find the *path matrix* for the graph **G** in **Q 5a** using powers of the adjacency matrix **A.**

c) Traverse the graph G shown below in **breadth first order, depth first order** and construct the **breadth first** and **depth first spanning trees.** Start from node **A** [if last digit of your ID is *odd*] / **G** [if last digit of your ID is *even*]. **4 CO2 C3**



The Graph G

بِسْمِ ٱللَّهِ ٱلرَّحْمَٰنِ ٱلرَّحِيمِ

International Islamic University Chittagong

Department of Computer Science and Engineering

BSc in CSE, Final Examination, Spring-2025

Course Code: **CSE-2323** Course Title: **Digital Logic Design**

Time: **2 Hours and 30 minutes** Marks: **50**

*[Answer **all** the following questions from Group A and B. Some questions may have an option. Use separate answer script for Group A & B. Figures at right margin illustrate the marks (M), course learning objectives (CLOs), and bloom's taxonomy (DL)]*

No.	Group A [2 X 10] = 20 marks	M	CLOs	DL
Q1. a)	Explain the limitations of a simple 4-to-2 encoder. How does a 4-to-2 priority encoder resolve these limitations? Include the logic expressions and diagram for its outputs in your explanation.	5	2	A
OR	Design a multiplexer using the following Boolean Equation. $$F(A,B,C) = \sum(0,1,2,3,4,5,6,7)$$			
b)	What is a possible solution to the carry propagation delay in parallel adders?	5	2	U
Q2. a)	When both J and K inputs of a JK flip-flop are 1, it can lead to an unstable condition. How can this problem be resolved? Illustrate your answer with a timing diagram and a circuit diagram of the solution.	5	2	A
b)	The SR flip-flop becomes invalid when both inputs are 1. If you want to use this input combination while maintaining the same behaviour for the other three input cases, you need to convert this flip-flop to another type. Which flip-flop would you choose for the conversion, and provide the necessary conversion logic for this redesign?	5	2	A
OR	Describe the operation of master-slave D flip-flop with logic diagram and characteristic table.			

No.	Group B [3 X 10] = 30 marks	M	CLO	DL
Q3. a)	Design a *Serial In Parallel Out shift register* and explain its operation.	5	1	U
OR	What is a memory cell? Design a 4x4 RAM and describe its operation with example			
b)	A combinational circuit is defined by the following functions: $F1(A,B,C) = \sum(3,5,,7)$; $F2(A,B,C) = \sum(4,5,7)$. Implement the circuit with a PLA having 3 inputs, 3 product terms and two outputs.	5	2	E
Q4. a)	A combinational circuit 3 function and defined by the following Boolean Expressions: $F_1 = x'y'z' + y'z$; $F_2 = xy'z' + x'y$; $F_3 = x'y' + xy'z$. Implement the circuit with a PLA having three inputs, Five product terms, and 3 outputs	5	3	A
OR	Implement the same combinational circuit with ROM			
b)	Design a 3-bit Magnitude Comparator digital circuit that compares two 3-bit binary numbers and determines their relative magnitudes with appropriate examples.	5	2	A
Q5. a)	Design a synchronous counter using T flip-flops to follow the sequence: $0 \rightarrow 1 \rightarrow 3 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 0 \rightarrow \ldots$ Mention the design steps clearly, then solve step by step.	5	3	A
b)	Design a 3-bits up-down counter using T flip-flops with control input μ (μ=1 for up, μ=0 for down). Draw the logic circuit with derived T flip-flops in equation.	5	2	2